

ROHAN SAMUEL GANGAVARAPU

+1 (480) 743-2550 | rgangava@asu.edu | [linkedin.com/in/rohan-samuel-g](https://www.linkedin.com/in/rohan-samuel-g) | github.com/sammy-16

EDUCATION

Arizona State University — Tempe, AZ

Master of Science in Computer Science

Aug. 2023 – May 2025

CGPA: 3.89/4

Vellore Institute of Technology — Vellore, India

Bachelor of Technology in Computer Science and Engineering

July 2017 – June 2021

CGPA: 8.44/10

EXPERIENCE

Software Engineer (GSA)

Arizona State University

Aug. 2024 – Present

Tempe, AZ

- Constructed an innovative AR iOS ecosystem leveraging Apple's ARKit, UIKit and Foundation libraries to transform user drawings and voice commands into interactive 3D objects.
- Integrated GPT-4V with multi-modal feedback systems and error analysis techniques that boosted AI code generation.

Software Development Intern

Wisk Aero LLC (Boeing Company)

June 2024 – Aug. 2024

Mountain View, CA

- Spearheaded the development of a distributed, multi-tiered embedded software systems in C/C++ enabling asynchronous communication between aerospace modules via Kafka, elevating system integration according to DO-178C aerospace certification standards.
- Formulated comprehensive quality assurance infrastructure by crafting pytest scripts achieving 100% code coverage.
- Configured VectorCAST for binary executable analysis, and built a custom Python Lizard-based cyclomatic complexity analyzer into Jenkins CI/CD pipelines for automated code quality enforcement.

Software Engineer Specialist

GE Aerospace

July 2021 – July 2023

Bengaluru, India

- Architected cloud-native web apps using Vue.js, React, and Java Spring Boot on event-driven architecture and RESTful APIs.
- Implemented infrastructure as code through Terraform and Kubernetes on AWS that dynamically scaled to support 5K+ daily users while fine-tuning performance through database indexing and pagination strategies.
- Delivered end-to-end software solutions by modeling lossless relational databases, prototyping high-fi Figma UI mockups.
- Orchestrated automated quality assurance through CI/CD pipelines with JUnit and Mockito, while fortifying security practices including SAST and DAST vulnerability scanning.
- Led development team as Scrum Master while driving 80% reduction in support ticket backlog (from 60+ to fewer than 10).
- Troubleshooting business-critical performance issues and receiving company 'Impact Award' for developing a single-page application that directly facilitated \$400M+ in business deals.

Software Engineer Intern

GE Vernova

Jan. 2021 – June 2021

Bengaluru, India

- Engineered full-stack renewable energy platform using Python Flask microservices and Angular, Node.js frontend.
- Optimized JSON processing, delivering 60% faster wind turbine assessments, 30% improved application efficiency, and 25% reduced data processing time, directly contributing to \$1B per unit revenue growth.

Data Analyst Summer Intern

Tech Trunk Ventures Private Limited

May 2019 – July 2019

Hyderabad, India

- Pioneered data analysis pipeline on machine learning models while automating data cleansing with NumPy scripts.
- Generated interactive visualizations using Python's Pandas, Seaborn and Matplotlib, achieving 85% accuracy in identifying optimal marketing seasons and dramatically enhancing decision-making capabilities.

PROJECTS

Dear | WebRTC, Hugging Face Transformers, MongoDB, Coqui XTTS, NLP

Present

- Developing an AI-powered voice fallback system for memory care patients by fine-tuning specialized transformer models on synthetic conversation datasets, implementing real-time voice recognition for elderly speech patterns, and developing an intelligent routing system that seamlessly transitions between AI simulation and actual family member connections.

LectureLlama - AI Agent | AWS - S3, Lambda, Bedrock, Python, Amazon Transcribe, Zoom AI Bot

Present

- Engineered an AI-powered classroom assistant using AWS Bedrock and Python that automatically transcribes Zoom lectures, provides real-time answers to student questions, detects and distributes important announcements via email, and post-lecture summaries reducing administrative workload, enhancing student engagement and information retention.

Fake it Till you Tweet it | Mistral AI, Python, Mixed-Regression, Research

Dec. 2024

- Engineered an AI-powered tweet generation system by fine-tuning Mistral language models and implementing a controlled experimental framework with varying prompt strengths (none/soft/hard), then conducted statistical analysis using mixed regression modeling to quantify human susceptibility to AI-generated content.

TECHNICAL SKILLS

Languages: Java, Python, C/C++, SQL (Postgres), JavaScript, Kotlin, R, C#, TypeScript, Rust, Go

Frameworks: Vue, Angular, React, Node.js, Flask, JUnit, Django, Material-UI, FastAPI, PyTorch, CUDA, DynamoDB

Developer Tools: Git, Docker, Kubernetes, Jenkins, Azure, GCP, AWS, VS Code, Postman, Spark, Redis, RedShift